

ECON 3640: PROBABILITY AND STATISTICAL INFERENCE

FALL 2024

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Office hours: By appointment

Class meetings: Monday and Wednesday, 1:25–2:45 PM, GC 2675 (section 001)

Prerequisites: College Algebra (MATH 1090), ECON 2010, and ECON 2020

Course Description

In this course, you will be exposed to statistical analysis through the lens of an economist. You will learn graphical and numerical methods of summarizing data, the concept of probability, and hypothesis testing. These skills are useful in a wide range of contexts, from reading and interpreting news articles critically and becoming an educated consumer to evaluating policies and preparing for advanced courses in quantitative analysis. This course belongs to the Quantitative Reasoning (QR-B) category and is a prerequisite for ECON 4650.

Learning Objectives

By the end of the course, you will be able to:

1. use R to organize, summarize, and visualize data and to carry out basic statistical analyses;
2. explain the logic of probability, sampling distributions, and statistical inference, and apply the most common procedures correctly;
3. distinguish correlation from causation in empirical data and in policy debates;
4. interpret and critique empirical results reported in economics journals and in the news.

Approach

Classes have a theoretical and an applied component. In theoretical lectures we develop the concepts and methods of each chapter; in applied lectures we carry out the same analysis in R, step by step, using the data sets from the textbook and economic data such as the Penn World Table. No previous knowledge of R is required. The problem sets and exams build directly on what we practice in class, so attending both kinds of lecture is the best preparation for them.

Required Materials

Layth C. Alwan, Bruce A. Craig, and George P. McCabe. *The Practice of Statistics for Business and Economics*, 5th edition. New York: Macmillan Learning, 2020.

The digital version of the book is available through the Bookshelf tab in Canvas (Inclusive Access program). It is also available from the Campus Store and the University Library. An older edition is acceptable, but the order of chapters and some content may differ.

Software. Applied work uses R, a free and open-source statistical programming language (<https://>

cloud.r-project.org). In applied lectures I use RStudio (<https://posit.co/download/rstudio-desktop>), which is also free; you may use another platform if you prefer. A spreadsheet editor such as Excel is recommended; University of Utah students have free access to Microsoft Office through Office 365.

Assessment

Problem sets (six, 5% each)	30%
Midterm exam	30%
Final exam (comprehensive, take-home)	40%

Problem sets. A new problem set is posted on Canvas whenever enough material has been covered in class. Answers copied directly from the lecture notes or from someone else's work receive no credit.

Exams. The midterm covers descriptive statistics and the first probability topics, with both theoretical and applied questions. The final exam covers all topics; it is take-home, and we will agree on a window that gives every student time to complete it. Exams closely reflect what you practice in class and in the problem sets.

Late work. Late problem sets are accepted up to 24 hours after the due date with a 20% penalty; submissions more than 24 hours late receive no credit. Equipment failures are not an acceptable excuse, so please start early enough to leave time for the unexpected.

Grading scale. A 93–100, A– 90–92.9, B+ 87–89.9, B 83–86.9, B– 80–82.9, C+ 77–79.9, C 73–76.9, C– 70–72.9, D+ 67–69.9, D 63–66.9, D– 60–62.9, E below 60 (percent).

Course Policies

Attendance is not part of the grade, but you are strongly encouraged to attend. Please be respectful of others; if you need to take a call, you are welcome to step out for a moment. I reply to email within 24 hours on weekdays and 48 hours on weekends and holidays; if you do not hear back within that time, please write again. If an emergency prevents you from taking an exam or completing the course, please let me know as soon as possible. Make-up exams are given only for a documented medical emergency or a conflict with another exam.

Course Schedule

The schedule is tentative. Chapter and section numbers refer to Alwan, Craig, and McCabe (2020).

Week	Topics	Notes
1	Introduction. R and RStudio; loading packages; organizing data with the tidyverse (Penn World Table)	
2	Examining distributions (1.1–1.4): types of variables; graphs; describing distributions with numbers; the Normal distributions	Problem set 1
3	Examining relationships (2.1–2.5): scatterplots; correlation; least-squares regression; cautions about correlation and regression; two-way tables	Labor Day (Sep 2)
4	Producing data (3.1–3.2): sources of data; designing samples	Problem set 2
5	Producing data (3.3–3.4): designing experiments; data ethics	
6	Probability (4.1–4.2): randomness; probability models	Problem set 3
7	Probability (4.3): general probability rules; conditional probability; Bayes's rule	Midterm (Oct 3–4)
8	<i>Fall break</i>	
9	Random variables and probability distributions (5.1–5.4): means and variances; binomial and Poisson distributions; continuous distributions	
10	Sampling distributions (6.1–6.3): sample means; the central limit theorem; sample proportions	Problem set 4
11	Introduction to inference (7.1–7.4): confidence intervals; tests of significance; use and abuse of tests; prediction intervals	
12	Inference for means (8.1–8.3): one-sample and matched pairs <i>t</i> procedures; comparing two means; sample size and power	Problem set 5
13	Inference for proportions (10.1–10.2): a single proportion; comparing two proportions	
14	Inference for categorical data (11.1–11.2): two-way tables; goodness of fit	
15	Nonparametric tests (17.1–17.3): Wilcoxon rank sum and signed rank tests; the Kruskal–Wallis test	Thanksgiving; problem set 6
16	Review	Classes end Dec 5
	<i>Final exam</i>	Dec 9–10

Other dates. Classes begin August 19; last day to add or drop, August 30; last day to withdraw, October 18; Thanksgiving break, November 28 to December 1.

University Policies and Resources

Updated mandatory syllabus policies regarding the Americans with Disabilities Act, Safety at the U, Addressing Sexual Misconduct, and Academic Misconduct can be viewed at <https://cte.utah.edu/instructor-education/syllabus/institutional-policies.php>.

Academic misconduct. It is expected that students adhere to University of Utah policies regarding academic honesty, including but not limited to refraining from cheating, plagiarizing, misrepresenting one's work, and/or inappropriately collaborating. This includes the use of generative artificial intelligence (AI) tools without citation, documentation, or authorization. Any student who engages in academic dishonesty may be subject to academic sanctions as per the University of Utah's Student Code: <https://regulations.utah.edu/academics/6-410.php>.

Drop and withdrawal. Students may drop a course within the first two weeks of a given semester without any penalties. Students may officially withdraw (W) from a class or all classes after the drop deadline through the midpoint of a course. A "W" grade is recorded on the transcript and appropriate tuition/fees are assessed. The grade "W" is not used in calculating the student's GPA. See the University's Academic Calendar for deadlines.

Campus resources. Center for Disability & Access, 162 Union Building, 801-581-5020, <https://disability.utah.edu>; campus police and SafeU, 801-585-COPS (801-585-2677), <https://safeu.utah.edu>; Office of Equal Opportunity and Affirmative Action (Title IX), 383 South University Street, 801-581-8365; Center for Student Wellness, 426 SSB, 801-581-7776, <https://wellness.utah.edu>; Basic Needs Collective, <https://basicneeds.utah.edu>; Learning Center, <https://learningcenter.utah.edu>; Writing Center, <https://writingcenter.utah.edu>; other student support, <https://studentsuccess.utah.edu/resources/student-support/>.

Syllabus changes. This syllabus is an outline and guide for the course. I may modify it, including the schedule, with reasonable notice; any changes will be posted on Canvas under Announcements.